4.1 Waves and quantum behaviour

		Learning outcomes	Additional guidance
(a)	*Describe and explain:*		
	(i)	production of standing waves by waves travelling in opposite directions	Including graphical treatment HSW1, 2
	(ii)	interference of waves from two slits	
	(iii)	refraction of light at a plane boundary in terms of the changes in the speed of light and explanation in terms of the wave model of light	*M0.2*
	(iv)	diffraction of waves passing through a narrow aperture	HSW6
	(v)	diffraction by a grating	
	(vi)	evidence that photons exchange energy in quanta $E = hf$ (for example, one of light-emitting diodes, photoelectric effect and line spectra)	limitations of particle and wave models HSW7, 11 *M3.1, M3.2, M3.3, M3.4*
	(vii)	quantum behaviour: quanta have a certain probability of arrival; the probability is obtained by combining amplitude and phase for all possible paths	*M1.3*
	(viii)	evidence from electron diffraction that electrons show quantum behaviour.	HSW7, 11

(b) *Make appropriate use of:*

(i) the terms: phase, phasor, amplitude, probability, interference, diffraction, superposition, coherence, path difference, intensity, electronvolt, refractive index, work function, threshold frequency.

(c) *Make calculations and estimates involving:*

(i)	wavelength of standing waves	end corrections not required
(ii)	Snell's Law, $n = \dfrac{\sin i}{\sin r} = \dfrac{c_{\text{1st medium}}}{c_{\text{2nd medium}}}$	*M03, M0.6, M4.5*
(iii)	path differences for double slits and diffraction grating, for constructive interference $n\lambda = d\sin\theta$ (both limited to the case of a distant screen)	angles may be given in degrees or radians, the use of the small angle approximation is expected. *M0.6, M4.6, M4.7*
(iv)	the energy carried by photons across the spectrum, $E = hf$	*M0.2, M2.3*
(v)	the wavelength of a particle of momentum p, $\lambda = \dfrac{h}{p}$	As given by the de Broglie relationship *M0.2, M2.3*

(d) *Demonstrate and apply knowledge and understanding of the following practical activities (HSW4):*

(i)	using an oscilloscope to determine frequencies	links to **4.1a(i)** **PAG5**
(ii)	determining refractive index for a transparent block	links to **4.1c(ii)** **PAG6**
(iii)	superposition experiments using vibrating strings, sound waves, light and microwaves	links to **4.1a(i), b(i), c(i)** **PAG5**
(iv)	determining the wavelength of light with a double-slit and diffraction grating	links to **4.1a(ii), a(v), c(iii)** **PAG5**
(v)	determining the speed of sound in air by formation of stationary waves in a resonance tube	links to **4.1a(i), c(i)** **PAG5**
(vi)	determining the Planck constant using different coloured LEDs.	links to **4.1a(vi), c(iv)** **PAG6**